Advanced Computer-Aided VLSI System Design

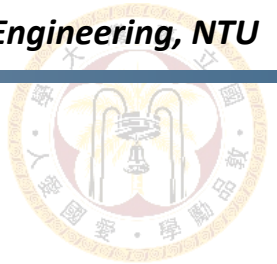
Chapter 9. Timing Signoff

Lecturer: Yueh-Feng Tsai

Graduate Institute of Electronics Engineering, National Taiwan University

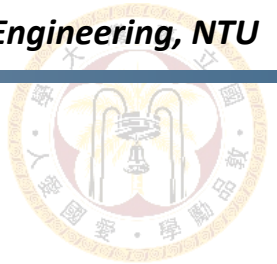


NTU GIEE



Outline

- Timing and PVT Variation Modeling
- Signoff Corner and OCV Guardband
- Post-route STA (DMSA)
- Timing ECO

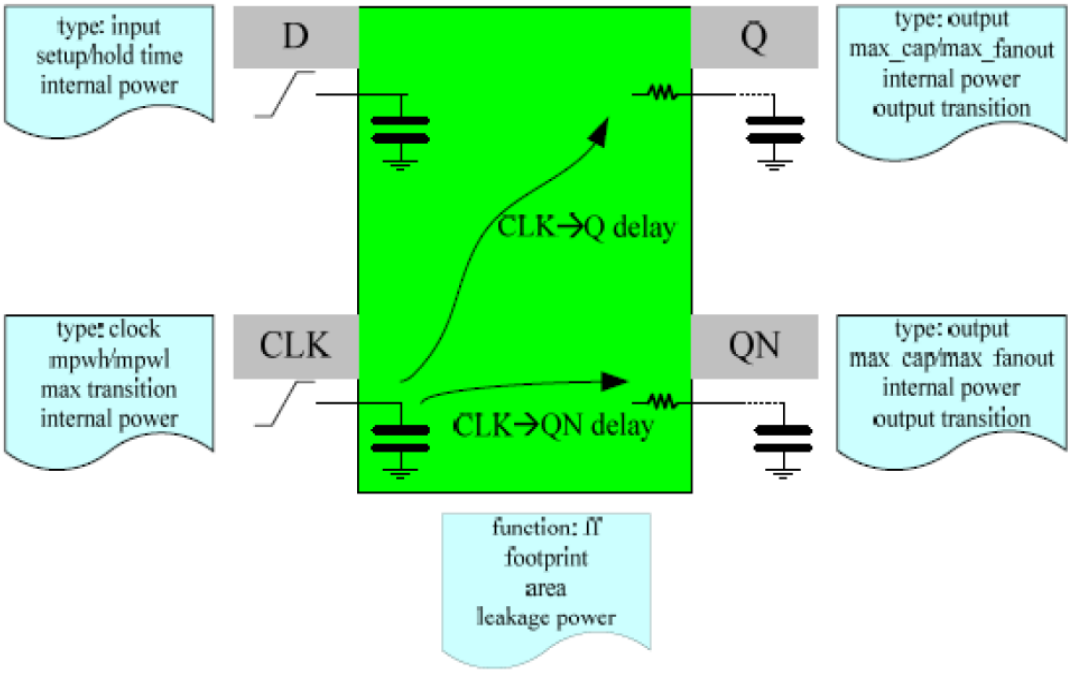


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- Signoff Corner and OCV Guardband
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- Timing ECO

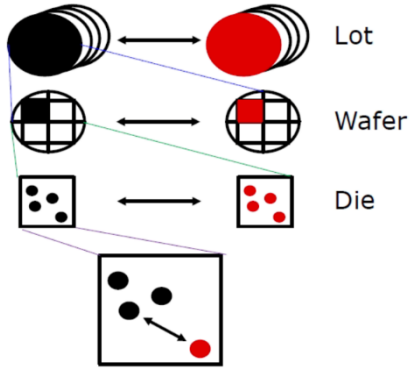
Liberty File (.lib)

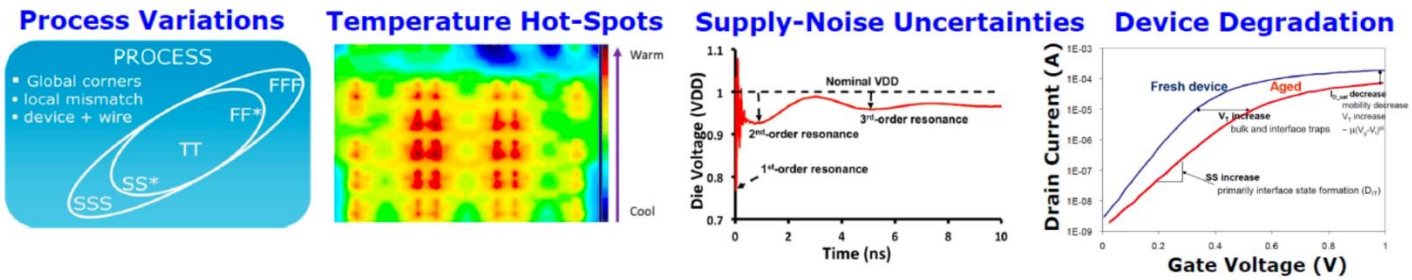
- **Operating conditions**
  - slow, fast, typical
- **Pin types**
  - input/output/inout
  - function
  - data/clock
  - capacitance
- **Path delay/transition**
- **Power**
- **Signal integrity**
- **Timing constraint**
  - setup, hold…



On-Chip Variation (OCV)

- **Process variation** (P)
  - Global mismatch of device driving strength
  - Local mismatch
  - Clock uncertainty, wire delay
- **Operating uncertainty** (V, T)
  - IR drop, temperature hotspot
- **Aging effects**
  - Electromigration
  - Transistor/device stress



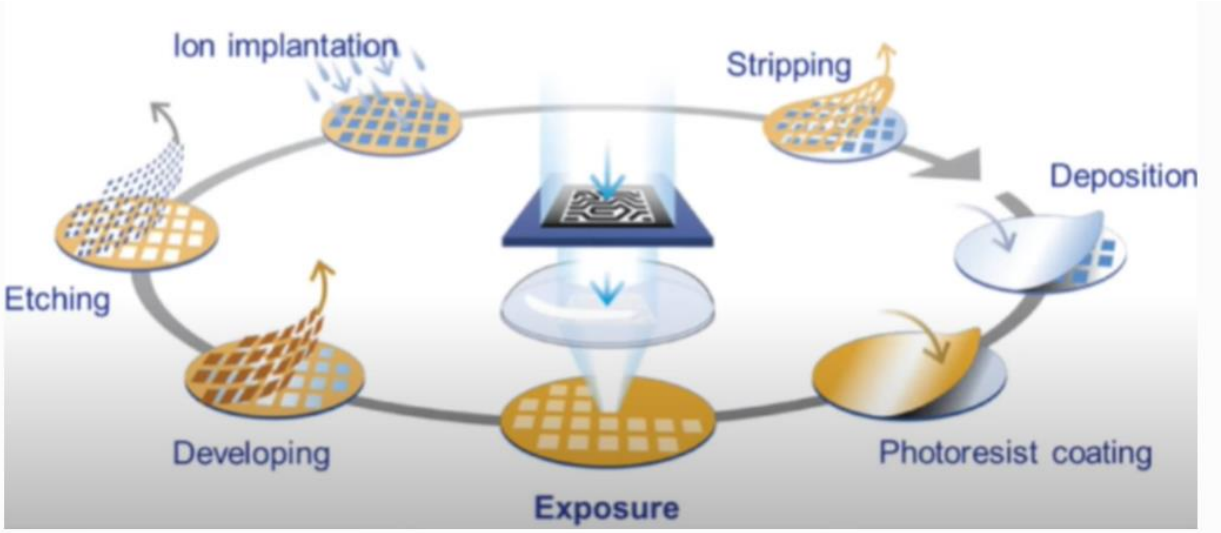


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J.-W Fang, ISSCC Tutorial 2021

Process Variation - 1

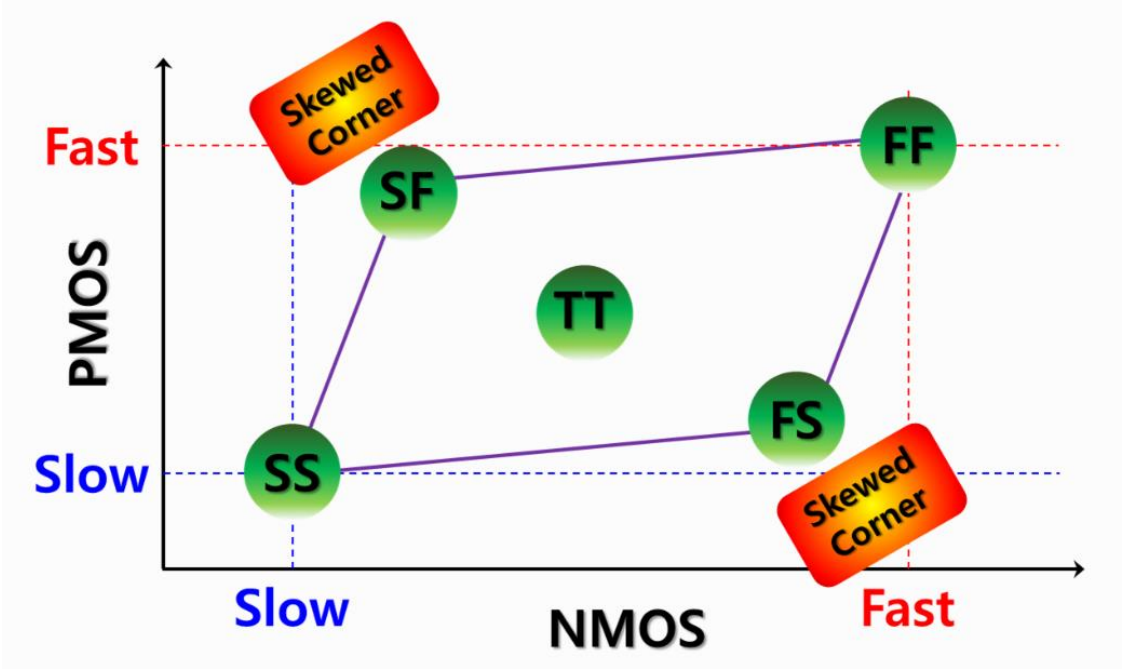
- During manufacturing, process variation arises from various technology/process steps



Source: What Are The Process Corners in VLSI Design - 딴딴's 반도체사관학교

Process Variation - 2

- Foundries provide skewed global corner in SPICE model to model the variation of P/N MOSFET
- Typically, you can also order skewed wafer from foundries to analyze the yield for each corner



Source: What Are The Process Corners in VLSI Design - 딴딴's 반도체사관학교

Process Variation - 3

- Foundries also provide the RC corner for the variation of Resistors and Capacitances.

Source: What Are The Process Corners in VLSI Design - 딴딴's 반도체사관학교

- RC values affect on IR drop and timing signoff

FF_1.2V_125°C	SS_1.05V_M40C	TT_1.05V_M40C
C-BEST	C-BEST	C-BEST
C-WORST	C-WORST	C-WORST
RC-BEST	RC-BEST	RC-BEST
RC-WORST	RC-WORST	RC-WORST

Hold Time Critical

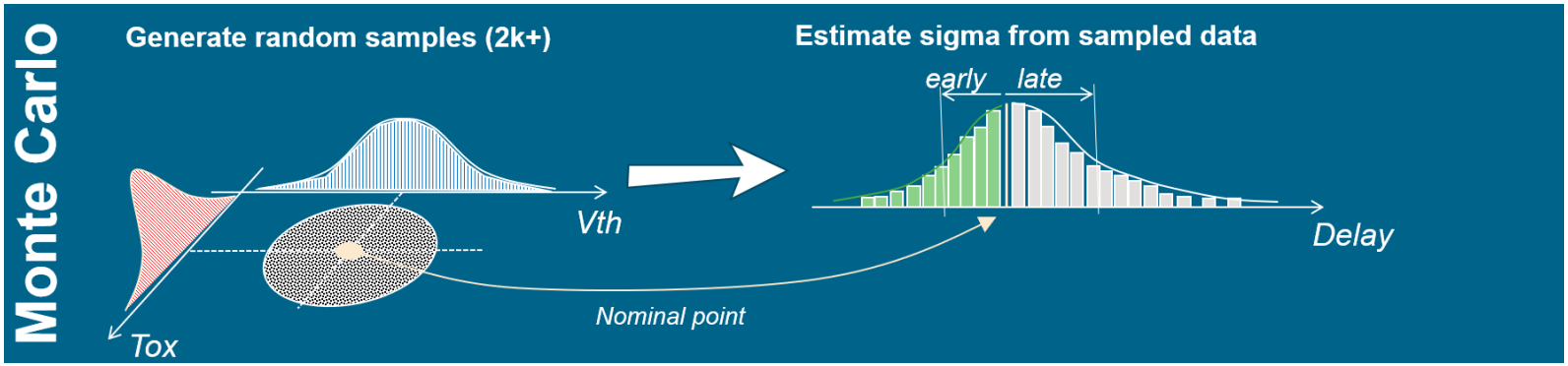
IR Critical

Setup Time Critical

Spice: Monte Carlo Simulation

- Monte Carlo simulation is widely used to find the distribution of process variation (Global+Local)
- Vth and oxide thickness variations cause device mismatch and logic-delay variation.

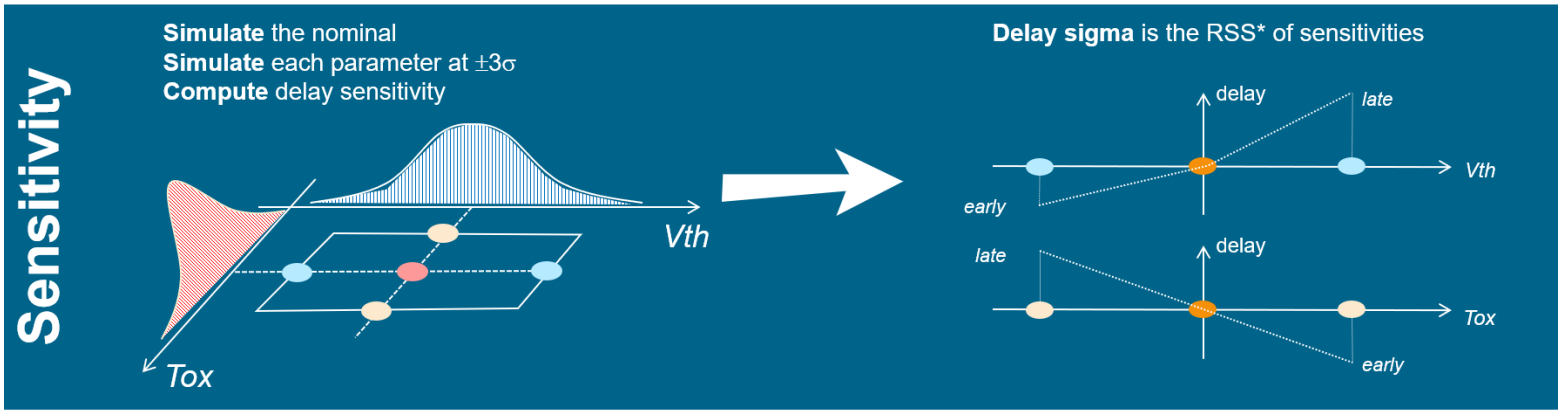
Source: Synopsys



STA: Sensitivity-Based Delay

- In advanced node technology, Liberty (.lib) files with variation information is widely used to consider on-chip variation.
  - Sensitivity table for delay, transition, load
  - Delay standard deviation (Sigma) is the root sum of squares of sensitivities (related to logic depth)

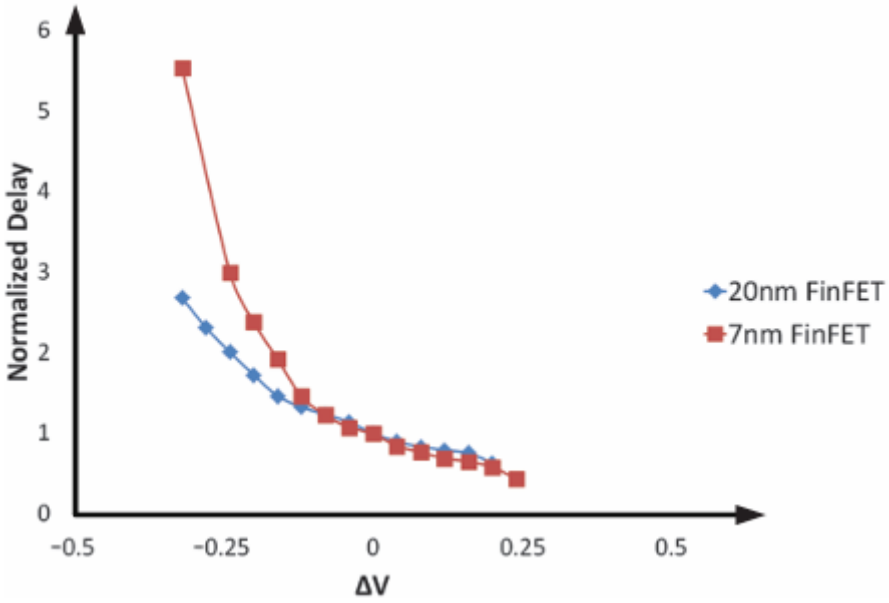
Source: Synopsys



*RSS – Root Sum of Squares

Voltage Variation

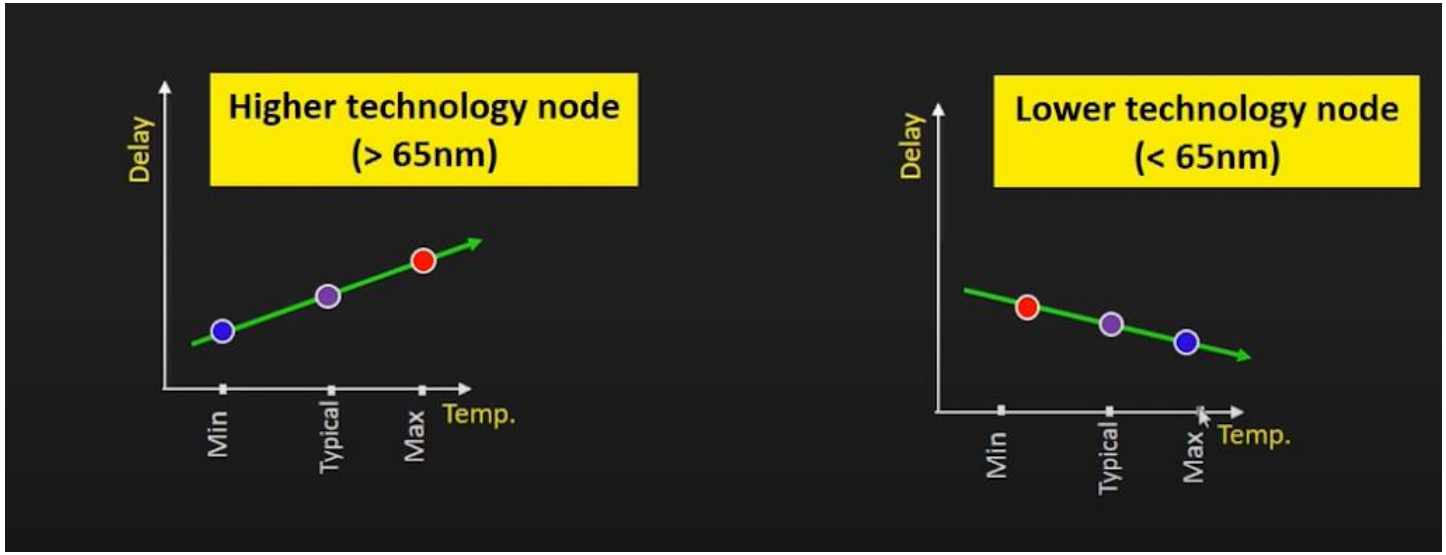
- Voltage affects standard cells' speed
- Dynamic Voltage Drop can affect the runtime timing variation of standard cells



Source: Paper

Temperature Variation

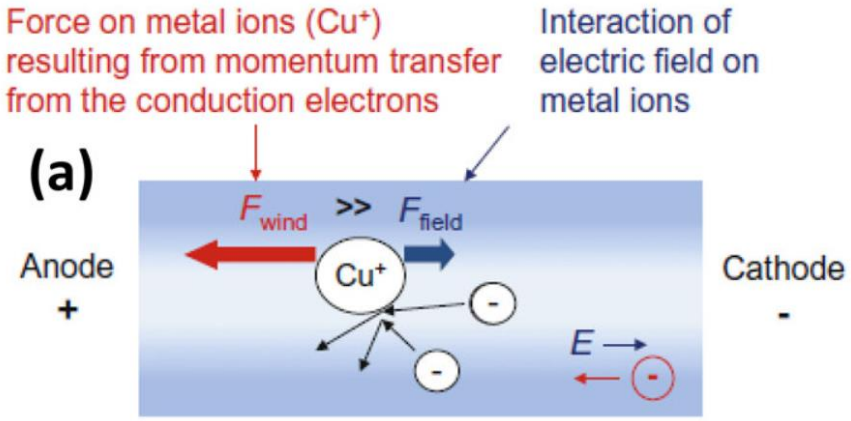
- Temperature-to-delay characteristic may have different trend (Temperature Inversion)
- In regional hotspot, standard cells could have lower delay and higher leakage current → Be aware of thermal gradient!

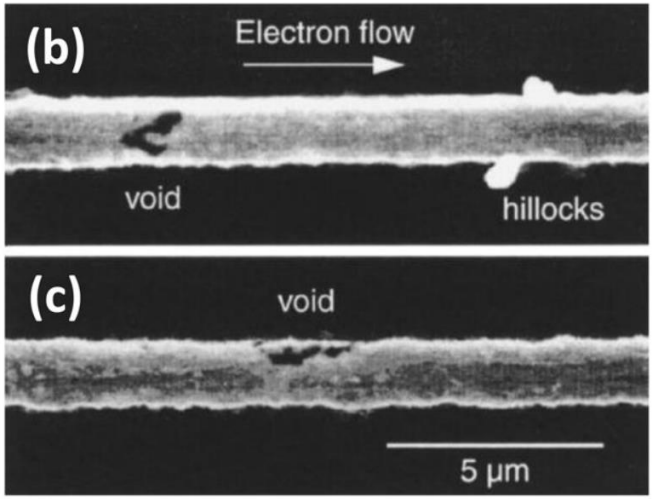


Aging Effect: Electromigration

- Metal atoms migrate under high current density, leading to void formation or wire breakage
- Electromigration degrades IC reliability and limits product lifetime over time
- Key influencing factors: current density, temperature, material properties, and interconnect geometry

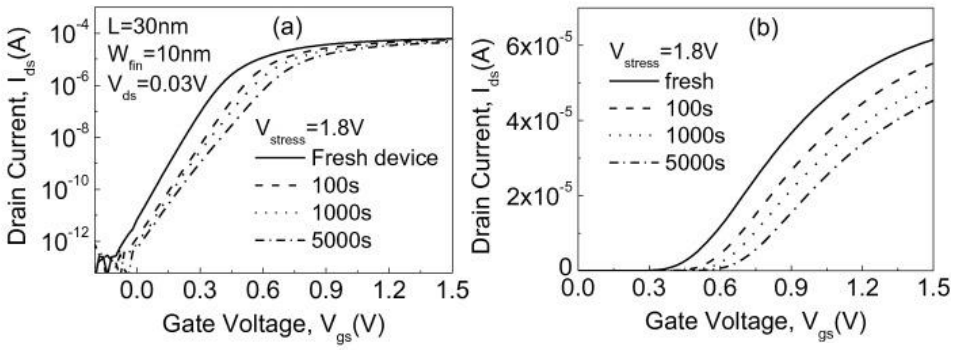
Source: Journal of Applied Physics



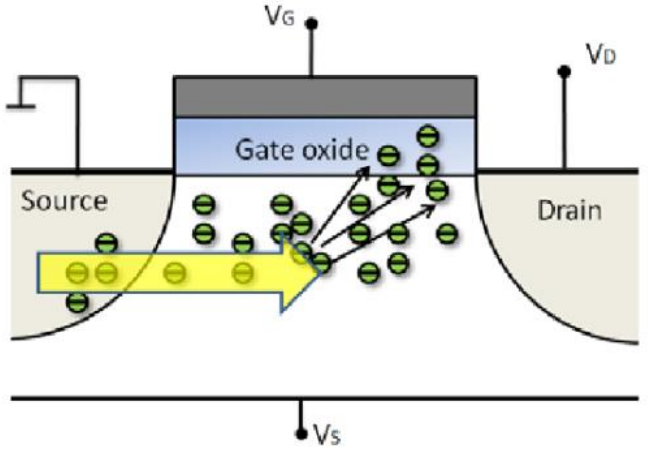


Aging Effect: Hot Carrier Effect

- Hot carriers are high-energy electrons or holes accelerated by a strong electric field near the drain
- These carriers can inject into the gate oxide, causing oxide damage and interface trap generation
- Primarily affects NMOS transistors under high drain-to-source voltage and low gate voltage conditions
- Results in threshold voltage shift, transconductance degradation, and reduced circuit speed



Source: Microelectronics Reliability

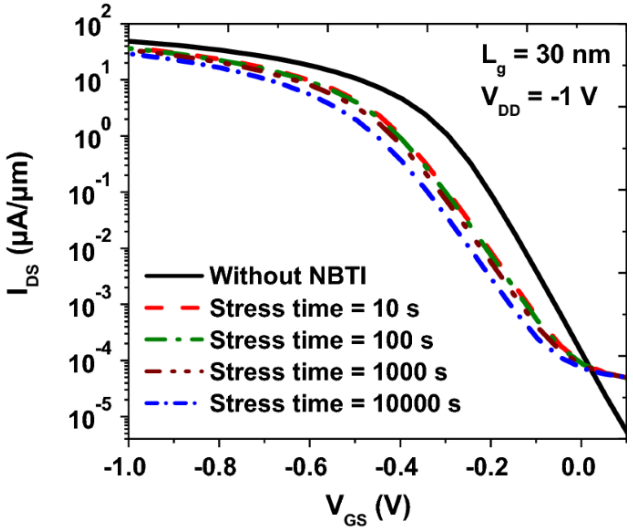


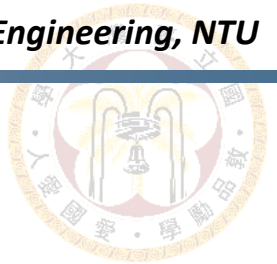
Source: Semantic Scholar

Aging Effect: Negative-Bias Temperature Instability

- NBTI mainly affects PMOS transistors under negative Vgs at elevated temperatures
- Caused by generation of interface traps and charge trapping in the gate oxide
- Results in threshold voltage increase and drive current degradation over time

Source: Microsystems Technology



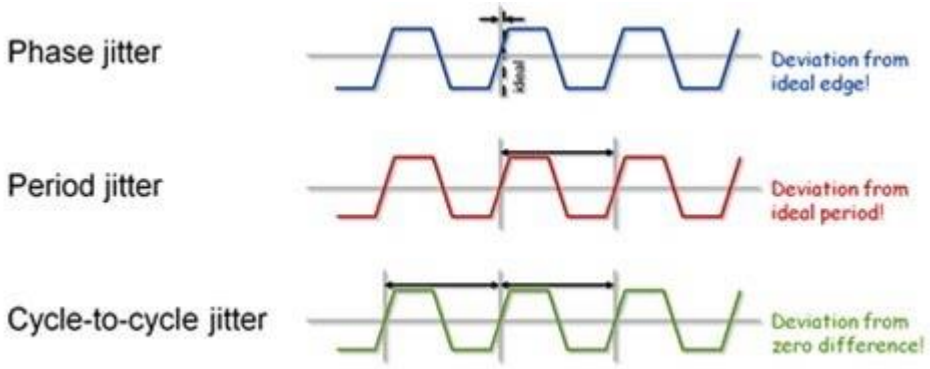


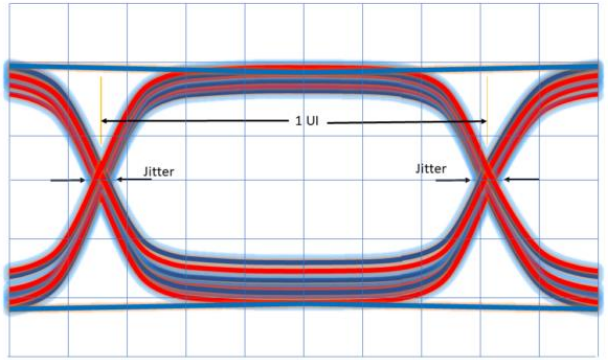
Summary of Aging Effect

Effect	Target Device	Trigger Condition	Impact	Mitigation
EM	Metal wire	High current	Open/short	Wider wire, Cu
HCI	NMOS	High Vds	Vth shift	Lower Vdd
NBTI	PMOS	High T, Vgs < 0	Vth ↑	Dynamic Vth ctrl

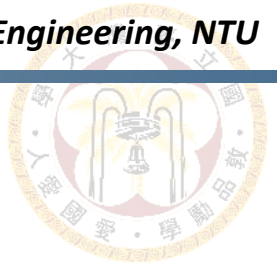
Time Variation – Jitter

- Jitter refers to the deviation in signal timing from its ideal position
- It introduces uncertainty in clock or data arrival times
- Causes include power supply noise, thermal fluctuations, and crosstalk
- Can be classified as deterministic or random jitter
- Excessive jitter can lead to timing violations and system instability





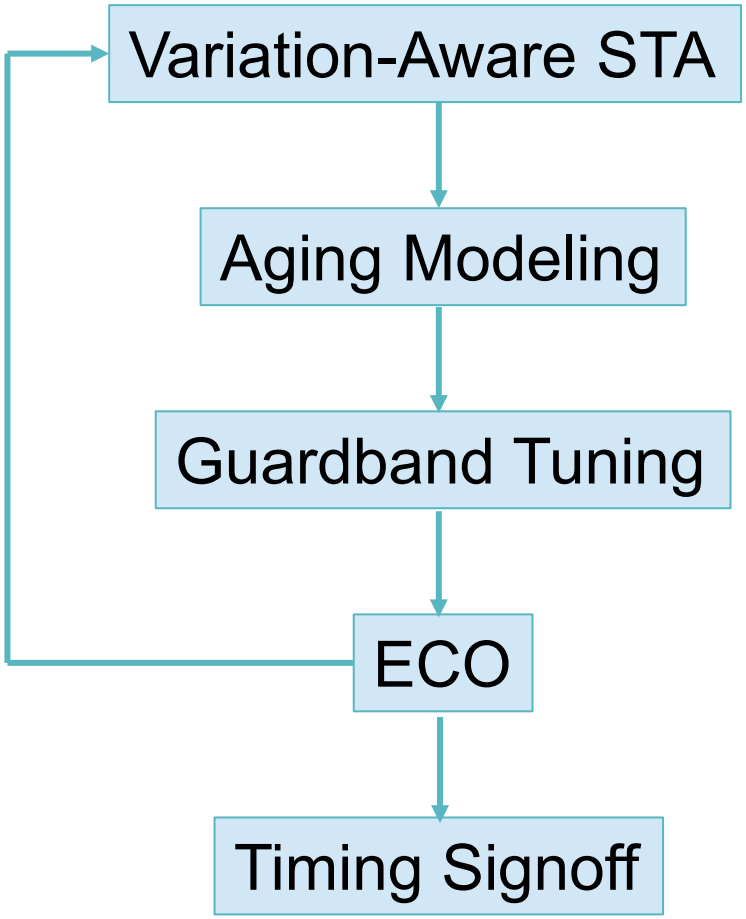
Source: Jitter



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Timing Signoff with PVTA Variation



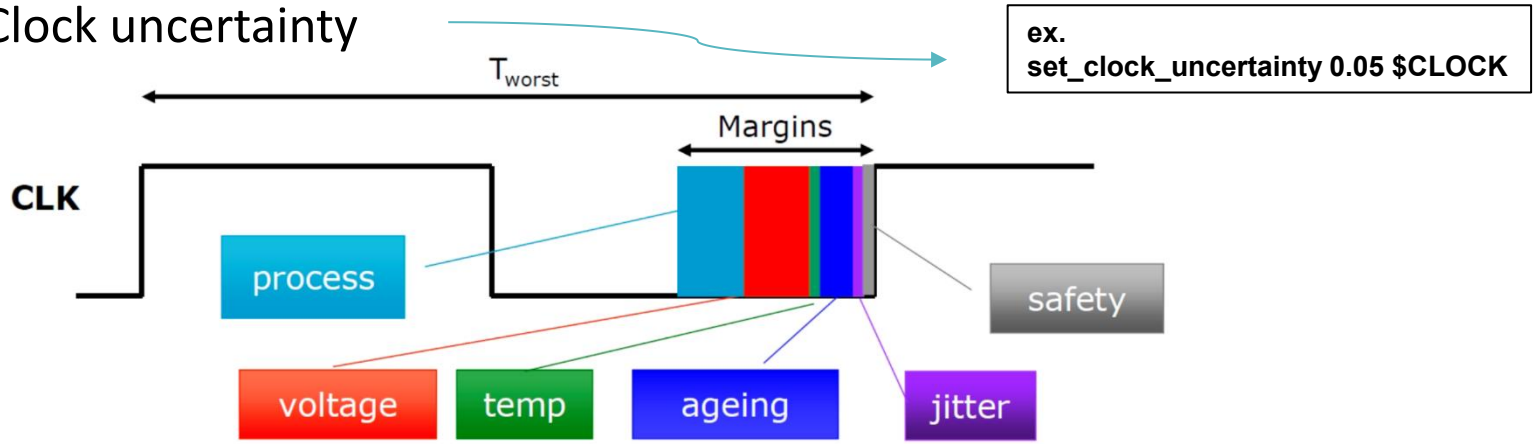
- RC SPEF File include
- SI Library Include
- OCV Setting (by side file or LVF)

- Foundry reference signoff guide (Setup/Hold/Leakage…)
- Add target sigma
- Add uncertainty for timing variation
- Clock Path Pessimism Removal

- ICC2+PrimeTime Timing ECO
- Innovus+Tempus Timing ECO

On-Chip Variation (OCV) Guardband

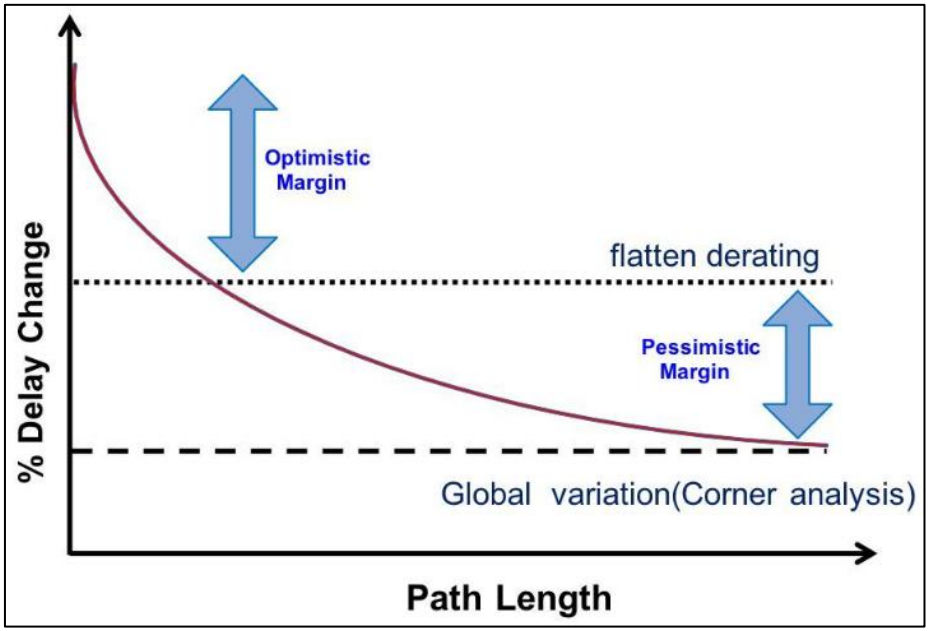
- **Timing signoff** is a challenging problem in advanced-node IC design
- Margin for the worst-case variation conditions limits F_{max} and V_{min}
- Related constraints to improve productivity
  - OCV, AOCV, POCV, Moment-based OCV
  - LVF flow (Liberty with variation adjustment)
  - Clock uncertainty



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OCV (Traditional Timing Derate)

- OCV : one derate value for all cells in one path, one corner



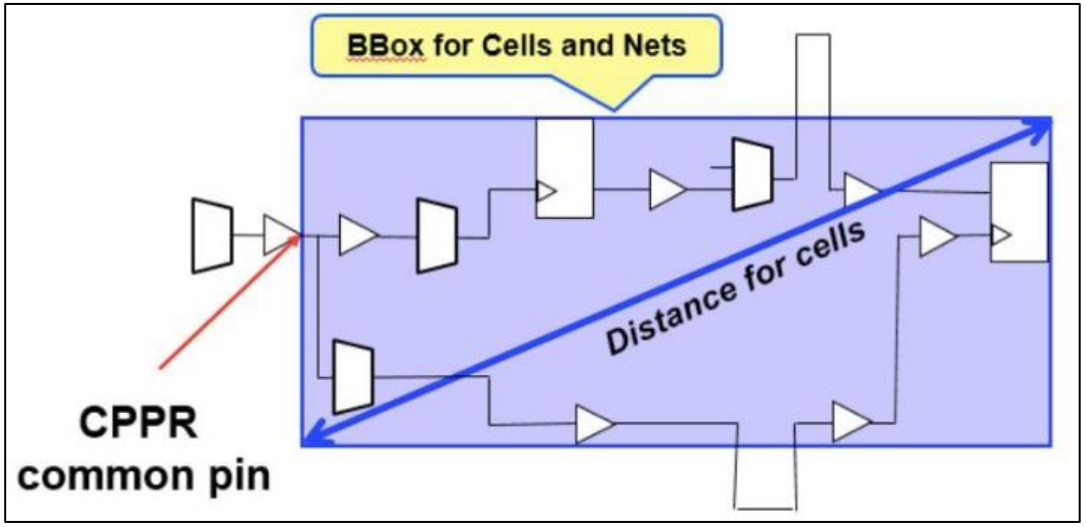
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ex.

```
set_timing_derate -early 0.9
set_timing_derate -late 1.1
```

AOCV (SBOCV)

- Advanced OCV (AOCV) or stage-based OCV (SBOCV) use the path depth and distance based derate
- Pros: More precise than OCV
- Cons: Does not include transition and load variation



1-d AOCV table :

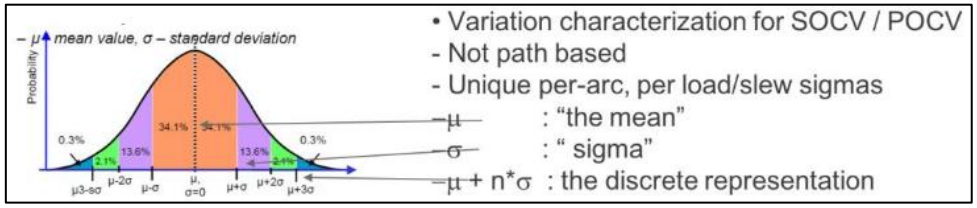
```
object_type : lib_cell
rf_type : rise
delay_type : cell
derate_lib_cell
rf_type : ristype : late
path_type : data
object_spec : 10nmlib/BUF_X4
depth : 1 2 3 4 5
table: 1.183 1.145 1.122 1.109 1.0901
```

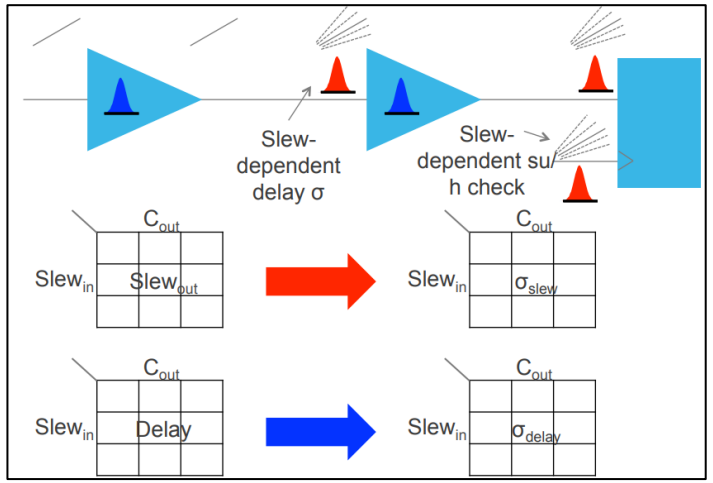
2-d AOCV table :

```
object_type : lib_cell
delay_type : cell
derate_type : early
path_type : data
object_spec : 10nmlib/BUF_X4
depth : 1 2 3 4 5
distance: 0 500 1000 1500
table: 1.123 1.090 1.075 1.067 1.062
       1.124 1.0911.076 1.068 1.063
       1.125 1.0921.077 1.070 1.065
       1.126 1.0941.079 1.072 1.067
```

POCV (SOCV)

- Parametric OCV (POCV) : Gaussian distribution of u (mean) and σ (sigma) for each cell, each index. Build from SSTA simplified model of local variable
  - Two format for POCV
    1. POCV side file
    2. LVF
- LVF : liberty variation format, including sigma table for cell delay, output transition, and constraint for each cell, each arc

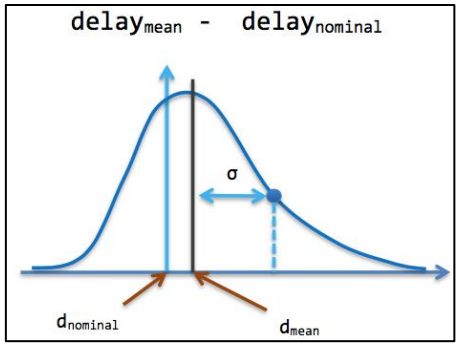




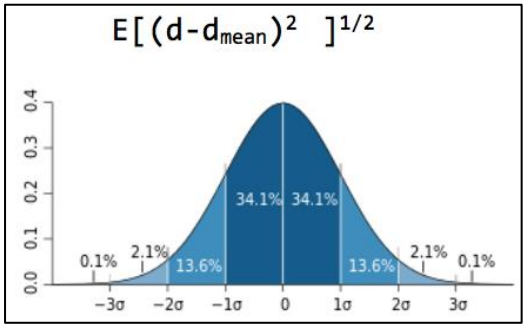
Moment-based LVF

- For strongly non-Gaussian distribution at ultra low voltage
  - Ex: Log-normal distribution
- Concerning mean-shift, standard deviation, and skewness

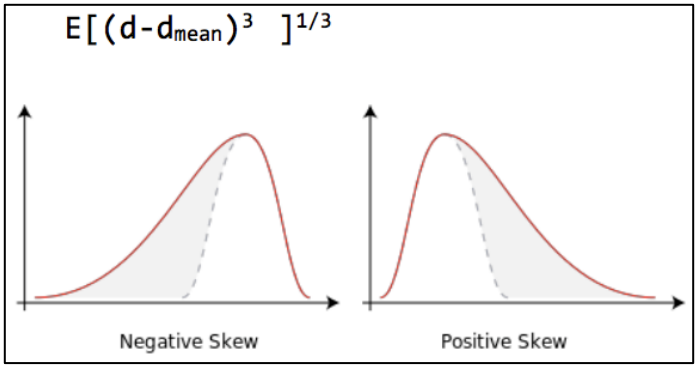
mean-shift



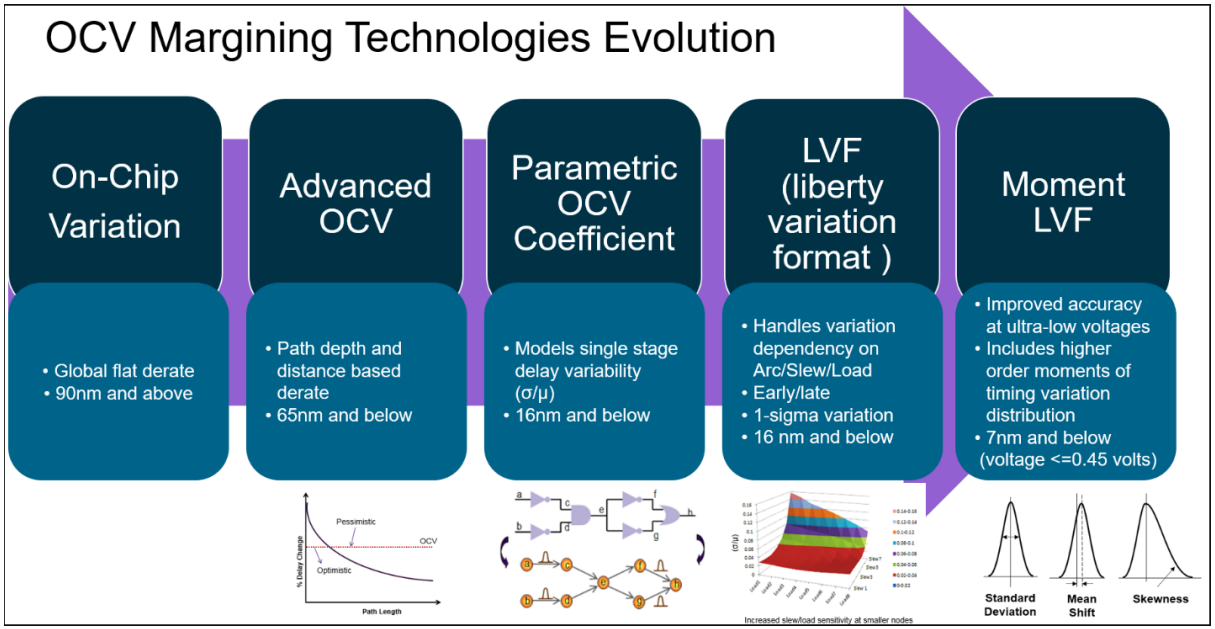
standard deviation

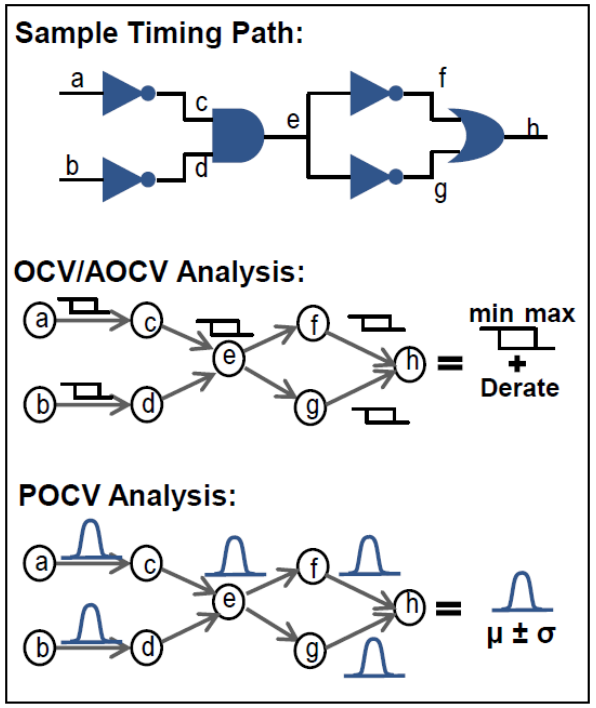


skewness



OCV Evolution Summary





PrimeTime STA with POCV

- Using POCV side file and LVF library

POCV Benefit	Side file	LVF
Eliminate GBA Pessimism	☑	☑
Improved P&R Correlation	☑	☑
Delay variation	☑	☑
Delay variation (slew/load dependent)		☑
Transition variation		☑
Constraint variation		☑
SiliconSmart Support	☑	☑

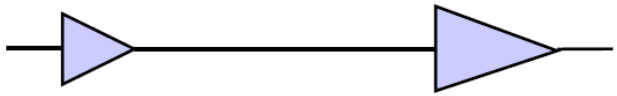
input data



pt setting

PrimeTime POCV – Path Calculation Example

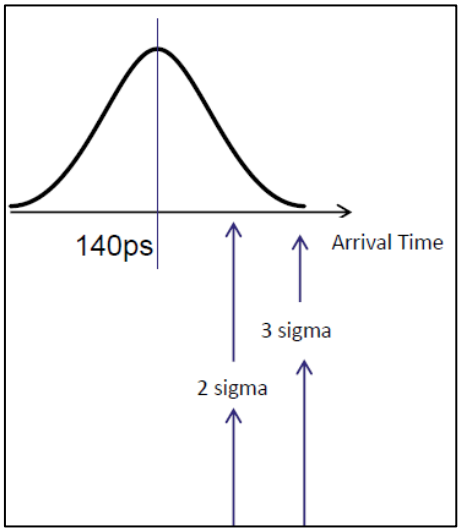
- Simple Root Sum Square (RSS)
  - Suggest both cells have POCVM coefficient of 0.05 or 5%



Nominal delay (mean) = 60ps Delay sigma = 0.05*60 = 3ps	Nominal delay (mean) = 80ps Delay sigma = 0.05*80 = 4ps

Path arrival nominal = 60+80 = 140ps
Path arrival sigma = √(3^2+4^2) = 5ps

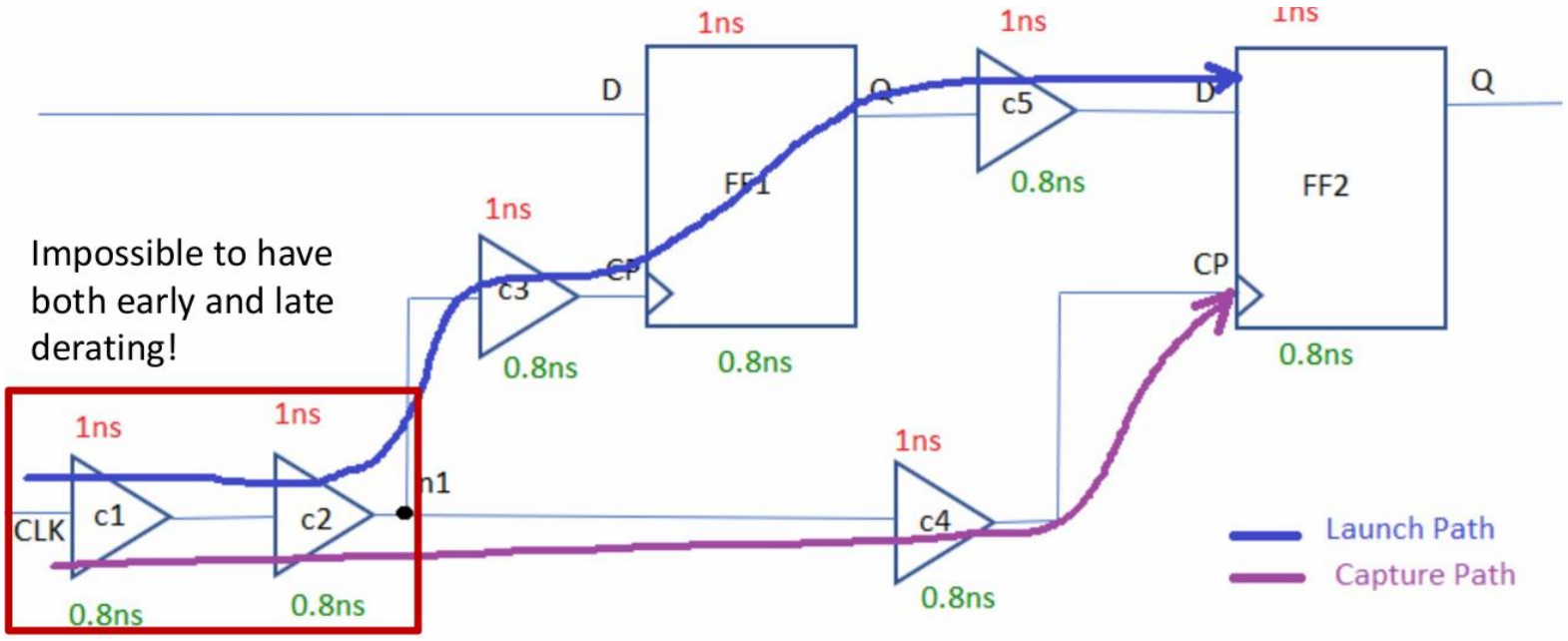
Path arrival at 2 sigma = 140+2*5 = 150ps
Path arrival at 3 sigma = 140+3*5 = 155ps



$$sigma_path^2 = sensit_cell_A^2 + sensit_cell_B^2 + sensit_cell_C^2 + \cdots$$

CRPR/CPPR for Reg2Reg Path

- Clock path pessimism removal (Cadence)
- Clock reconvergence pessimism removal (Synopsys)
- Remove pessimistic timing analysis in the clock common path by OCV

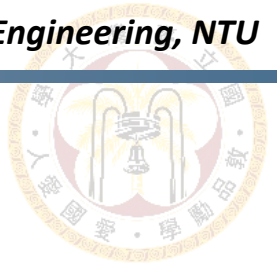


PrimeTime POCV – Constraint

- Constraint variation is supported in LVF formal
  - Setup and hold
  - Recovery and removal
  - Non-sequential setup and hold
  - Nochange setup and hold
  - Clock-gating check
- Using input transitions of constraint pin and related pin, constraint sigma will be calculated using LVF data in the library
- Constraint sigma is added into sigma of `data required time`
- Constraint variation is off by default. To enable it
  - `pt_shell> set_app_var timing_enable_constraint_variation true`

PrimeTime POCV – Transition Variation

- Transition variation is supported in LVF format
  - Improved accuracy in Monte Carlo correlation
  - Has no effect if POCV side file is used as an input for delay variation
- Transition variation is folded into cell delay variation
  - Reporting commands still report **nominal** transition. With and without transition variation, reported delay variation will be different
  - Based on input transition and output load, output transition sigma will be calculated using LVF tables in the library
- Output transition sigma will be propagated to the input of receiver cell
- Transition variation is off by default. To enable it
  - `pt_shell> set_app_var timing_enable_slew_variation true`

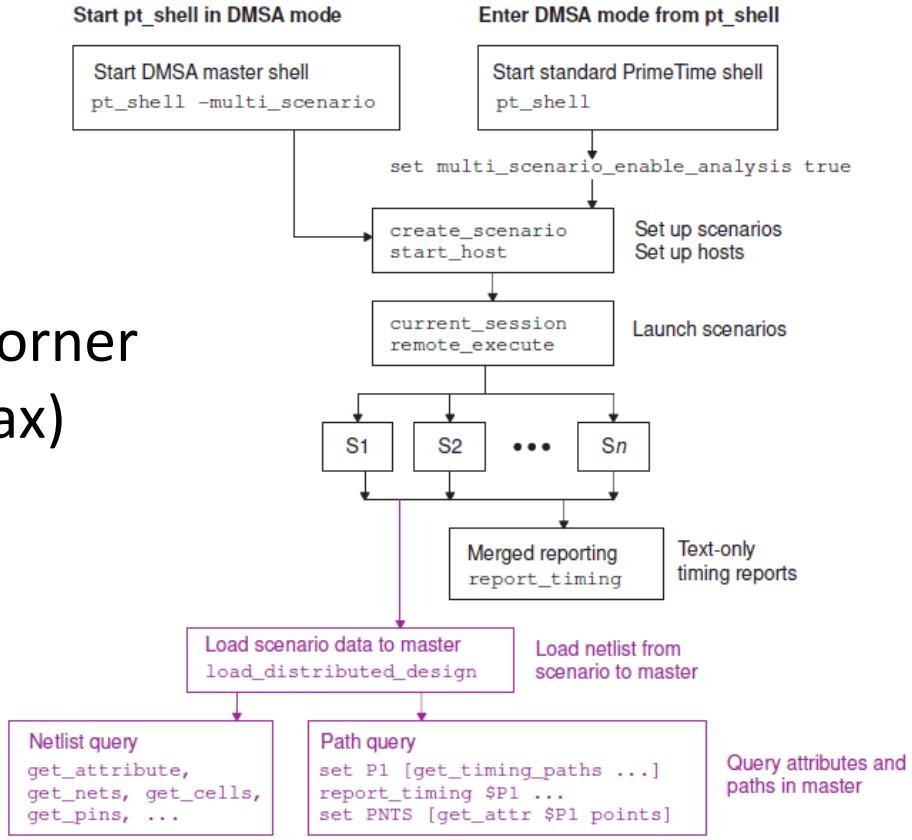


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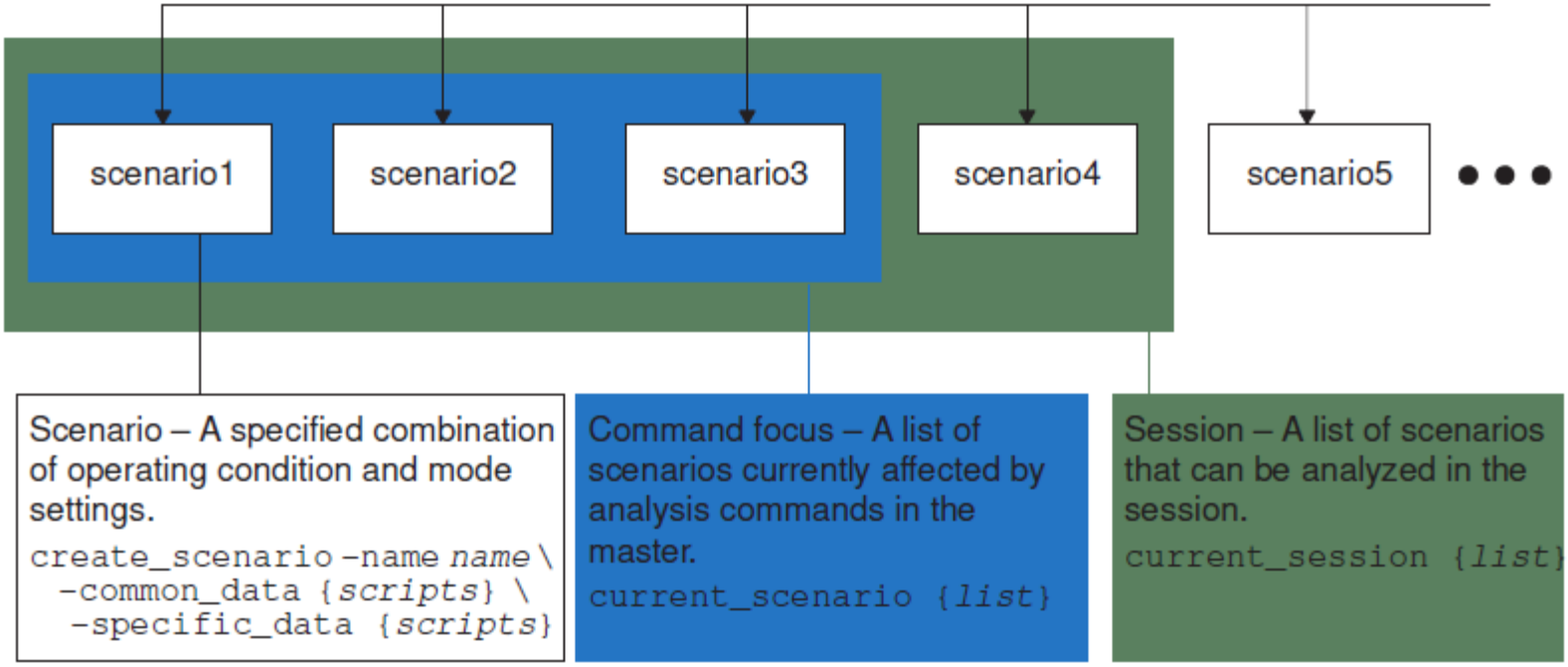
PrimeTime DMSA Flow

- Distributed Multi-Scenario Analysis enable timing analysis in different scenarios on different host
- Enable DMSA mode
  - pt_shell –multi-scenario
- Scenario: function mode + corner
  (Func + ss0p72vm40c + rcmax)
  (DFT + ff0p88v125c + rcmin)



create_scenario

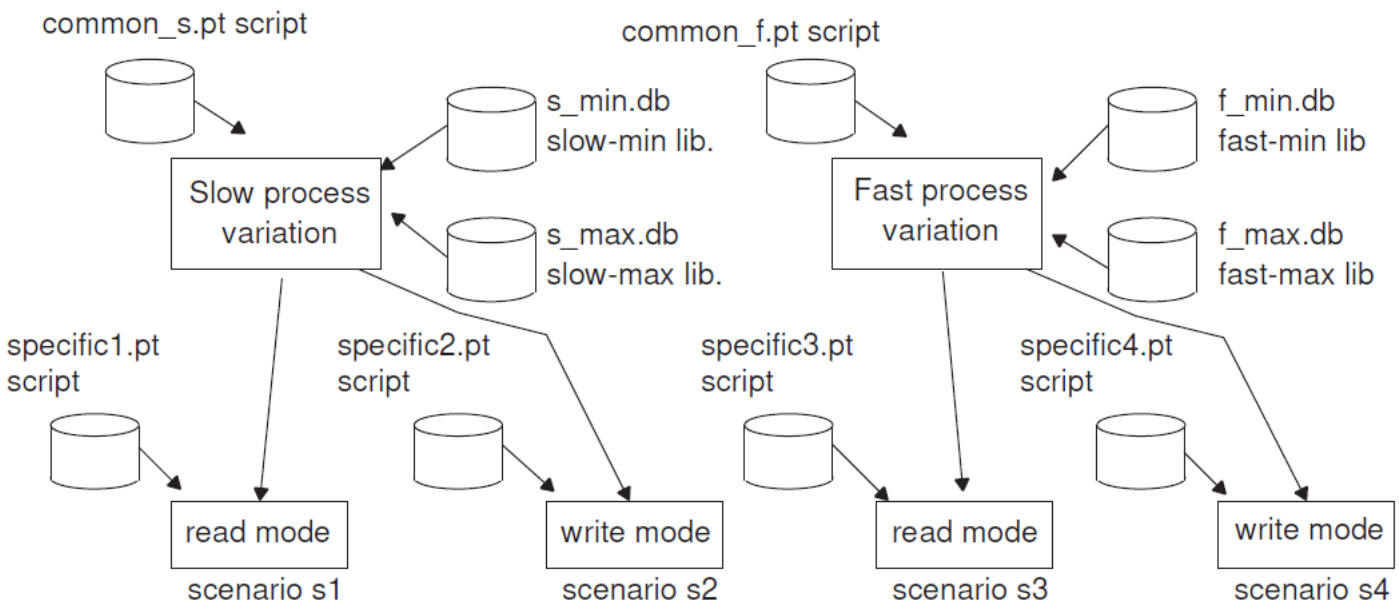
- create_scenario \
  - name $scenario_name \
  - common_data $common_sdc_file \
  - specific_data $specific_sdc_file



Create Scenario Example

Source: Synopsys PT 2022.2 UG

Figure 15 Multi-Scenario Setup Example



This design is to be analyzed at two process variation extremes, called slow and fast, and two operating modes, called read and write. Thus, the four scenarios to be analyzed are slow-read, slow-write, fast-read, and fast-write.

Configure Compute Resource

- set_host_options
  - num_processes # the number of processes per host
  - name # the name of the host option
  - max_cores # the number of CPU Cores per process
  - submit_command # specify the command that is used to start the PT (rsh (remote shell, by default))

In the following example, the host option is named ptopt030, rsh is used to connect to the host, and a maximum of three cores per process are specified for the compute resources:

```
pt_shell> set_host_options -name my_opts -max_cores 3 \
          -num_processes 4 -submit_command "/usr/bin/rsh -n" ptopt030
```

Specify the maximum number of cores by specifying the remote core count with the `set_host_options` command in the manager script, as shown in the previous example.

Source: Synopsys PT 2022.2 UG

Verify the Host Option

- report_host_usage: This command reports peak memory and CPU usage for the local process and all distributed processes that are already online.

```
pt_shell> report_host_usage
****************************************
Report : host_usage
...
****************************************

Options Name      Host              32Bit     Num Processes
-------------------------------------------------------------
my_opts           ptopt030          N         4

Usage limits (cores)
Options Name                                  #     Effective
-------------------------------------------------------------------------
(local process)                               -     2
my_opts5                                            -
-------------------------------------------------------------------------
Total                                               2

Memory usage
Options Name      #        Peak mem (MB)
-----------------------------------------
(local process)   -        24.93

Performance
Options Name      #          CPU time (s)            Elapsed Time (s)
-----------------------------------------------------------------------
(local process)   -          3                       25
```

Source: Synopsys PT 2022.2 UG

Start Hosts

- If you want to bring the host option online, you can use the "start_host" command

  after start_host and report_host_usage:

```
pt_shell> report_host_usage
...
Options Name       Host              32Bit     Num Processes
-------------------------------------------------------------
my_opts            ptopt030          N         4

Options Name    #     Host Name       Job ID     Process ID   Status
-------------------------------------------------------------------
my_opts         1     ptopt030        28857      28857        ONLINE
                2     ptopt030        28912      28912        ONLINE
                3     ptopt030        28966      28966        ONLINE
                4     ptopt030        29020      29020        ONLINE

Usage limits (cores)

Options Name                          #     Effective
-------------------------------------------------------------
(local process)                       -     2
                                      1     -
                                      2     -
                                      3     -
                                      4     -
-------------------------------------------------------------
Total                                       2

Memory usage

Options Name     #         Peak Memory (MB)
-----------------------------------------
(local process)  -         24.93
my_opts          1         26.31
                 2         26.31
                 3         26.31
                 4         26.31

Performance
```

Source: Synopsys PT 2022.2 UG

Change the scenarios in focus in the current session

- current_session {*list_of_scenarios*}

- current_session is used to set the scenarios to be analyzed

In the following example, a configuration and three scenarios named scen1, scen2 and scen3 are created. The current session is set such that scen2 and scen3 are in session and command focus.

- scen1 is created using common_s1.pt and specific_s1.pt
- scen2 is created using common_s2.pt and specific_s2.pt
- scen3 is created using common_s3.pt and specific_s3.pt

The scen2 and scen3 scenarios are brought into focus.

```
pt_shell> create_scenario -name scen1 -common_data {common_s1.pt} \
            -specific_data  {specific_s1.pt}
1
pt_shell> create_scenario -name scen2 -common_data {common_s2.pt} \
            -specific_data  {specific_s2.pt}
1
pt_shell> create_scenario -name scen3 -common_data {common_s3.pt} \
            -specific_data  {specific_s3.pt}
1
pt_shell> current_session {scen2 scen3}
{"scen2", "scen3"}
```

Select the Scenarios that are Command Focus

- current_scenarios {*list_of_scenarios*}
- In DMSA, after you **create scenarios**, use the **current_session** to focus on a set of scenarios for analysis. Next, focus the analysis on a specific subset of the scenarios in the current session by running the **current_scenario** command; subsequent commands will apply **only** to specified scenarios.

```
pt_shell> current_scenario {scen2 scen3}
{"scen2", "scen3"}
pt_shell> report_multi_scenario_design -session

****************************************
Report : multi_scenario_design
 ...
****************************************


Session details
-------------------------------------------------

Number of scenarios in current session: 3

    Focus          Scenario  (Image provider)

    Session        scen1   (scen1)
    Command        scen2   (scen2)
    Command        scen3   (scen3)
```

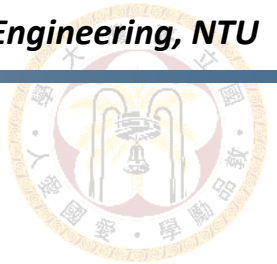
Executing Commands Remotely

- To explicitly run commands in the worker context, use the remote_execute command.
- Enclose the list of commands as a Tcl string. Separate the command with semicolons so that they execute one at a time.
- remote_execute
  - pre_command # the command before remote executed command
  - post_command # the command after remote executed command

In the next example, the situation is as in the first example except that the **-pre_commands** option is used to set a variable before the report.

```
pt_shell> remote_execute  -pre_commands {set
rc_slew_lower_threshold_pct_fall 30.0} {report_timing}

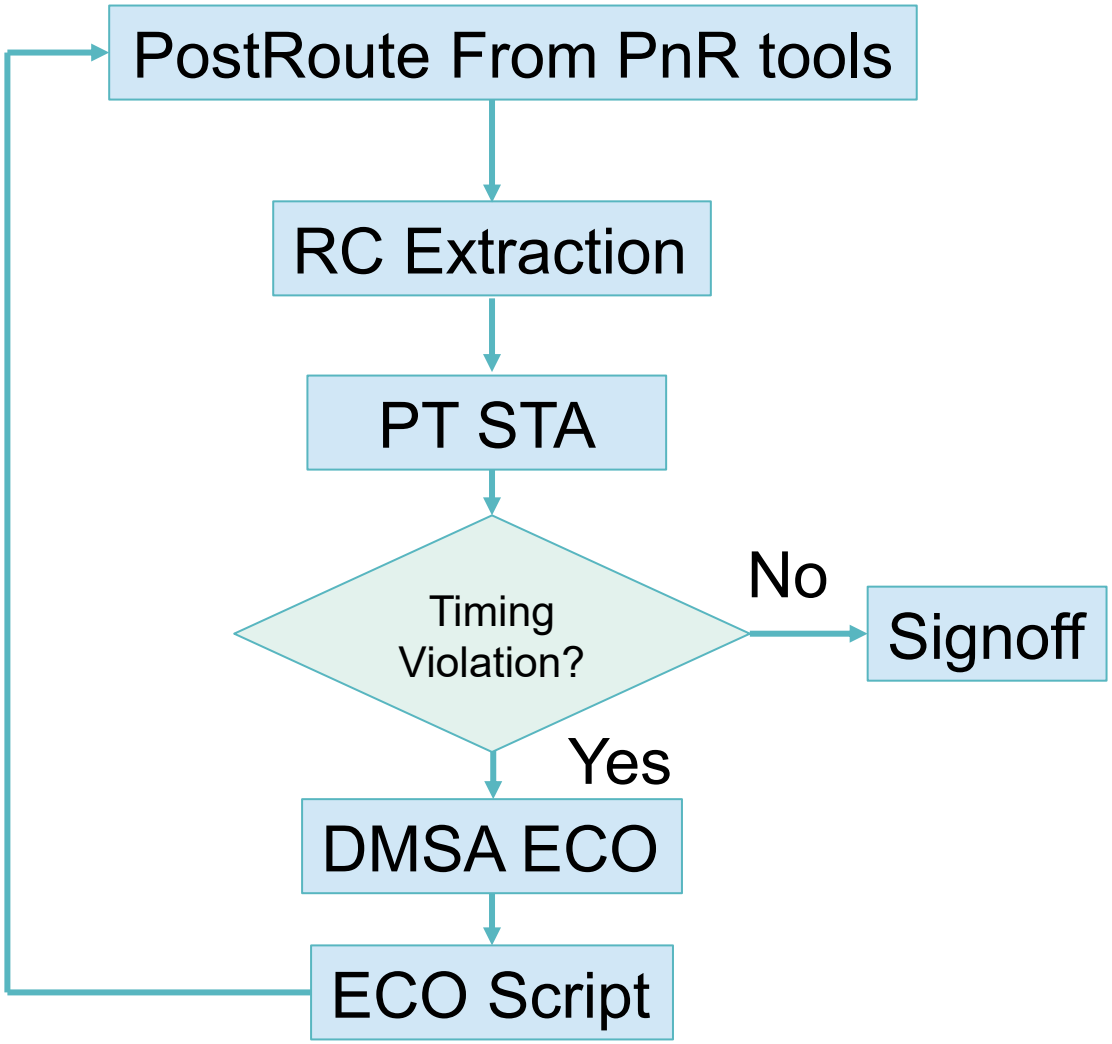
Start of Master/Slave Task Processing
-------------------------------------
Started  : Command execution in scenario 's1'
Command execution in scenario 's1' : Succeeded
Finished : Command execution in scenario 's1'
---------------------------------
End of Master/Slave Task Processing
1
```



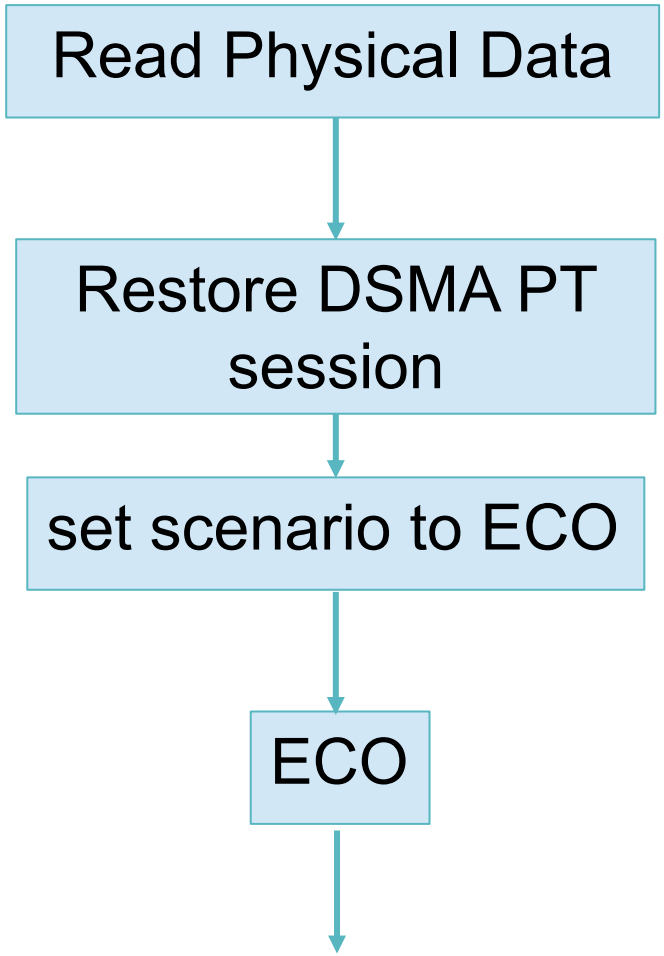
Outline

- Timing and PVT Variation Modeling
- Signoff Corner and OCV Guardband
- Post-route STA (DMSA)
- **Timing ECO**

Reference Flow with EDA tools



ECO with PrimeTime DMSA



- set_eco_options
  - Tech LEF
  - Macro LEF
  - DEF file

- restore_session $pt_session (Include many scenarios)
- Set current scenario for eco

- fix_eco_drc
- fix_eco_timing
- fix_eco_power
- fix_eco_leakage
- write_changes

set_eco_options

- There are many eco option can be applied in PrimeTime
- set_eco_options \ #If you use Innovus for PnR
  - physical_tech_lib_path $tech_lef_file
  - physical_lib_path $other_lef_files
  - physical_design_path $def_file
  - timing_setup_margin $target_setup_margin
  - timing_hold_margin $target_hold_margin
- Include physical data to improve timing correlation after eco between APR tools and PrimeTime
- set_eco_options \ #If you use ICC2 for PnR
  - physical_icc2_lib $icc2_lib_path
  - physical_icc2_blocks $icc2_blocks

Step 1: fix_eco_power

- fix_eco_power
  - dynamic_scenario
  - leakage_scenario
  - hold_margin (ex: 0.05)
  - setup_margin (ex: 0.05)
  - method (can be size_cell or remove_buffer)
  - pattern_priority (ex: {HVT RVT LVT})
  - cell_type (can be combinational, sequential or clock_network)
  - power_mode (can be total, dynamic or leakage)

Steps	Commands	Fixing mechanisms	Precedence rules
Step1: Power recovery	`fix_eco_power`	Cell sizing, buffer removal	Does not introduce new timing or DRC violations

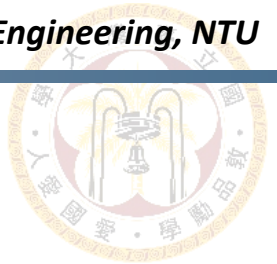
fix_eco_power Example - 1

- fix_eco_power should be added after set power_enable_analysis true

The following script performs area and power recovery in a distributed multi-scenario analysis (DMSA) flow.

```
#-------------------------------------------------------------------
# Update timing & global reporting in master
#-------------------------------------------------------------------
remote_execute {
  set timing_save_pin_arrival_and_slack true; update_timing -full
}
report_global_timing
report_constraint -all -max_capacitance -max_transition

#-------------------------------------------------------------------
# Before area/power recovery:
#   Report timing, power, cell usage for each worker process
#-------------------------------------------------------------------
remote_execute {
  report_global_timing
  report_constraint -all -max_capacitance -max_transition
  set power_enable_analysis true
  report_power -group {...}
  report_cell_usage
}
```



fix_eco_power Example - 2

```
#-------------------------------------------------------------------
# Before area/power recovery:
#   Check compatible library cells for ECO
#-------------------------------------------------------------------
remote_execute {
  report_eco_library_cells
}

#-------------------------------------------------------------------
# Execute signoff-driven area/power recovery.
#-------------------------------------------------------------------
fix_eco_power -verbose

#-------------------------------------------------------------------
# Reporting after area/power recovery
#-------------------------------------------------------------------
report_global_timing
report_constraint -all -max_capacitance -max_transition
remote_execute {
  report_global_timing
  report_constraint -all -max_capacitance -max_transition
  set power_enable_analysis true
  report_power -group {...}
  report_cell_usage
}
exit
```

Step 2: fix_eco_drc

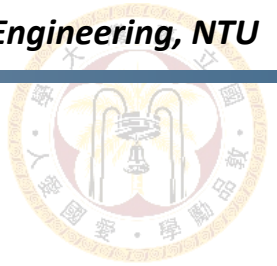
- fix_eco_drc
  - type (can be max_capacitance/transition/fanout, noise, delta_delay, cell_em)
  - hold_margin (ex: 0.05)
  - setup_margin (ex: 0.05)
  - method (can be size_cell or insert_buffer…)
  - cell_type (combinational, sequential or clock_network)

DRC, Crosstalk, and Cell Electromigration Violation Fixing

To fix design rule constraint (DRC), noise, crosstalk delay, or cell electromigration violations, use the `fix_eco_drc` command. It fixes the violations reported by the following commands:

```
report_constraint -max_capacitance ...
report_constraint -max_transition ...
report_constraint -max_fanout ...
report_noise ...
report_si_bottleneck ...
report_cell_em_violation ...
```

Source: Synopsys PT 2022.2 UG



Step 3: fix_eco_timing

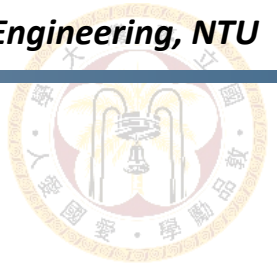
- If the STA report has negative slack on setup and hold time, you can use fix_eco_timing

- fix_eco_timing
  - type setup/hold
  - hold_margin (ex: 0.05)
  - setup_margin (ex: 0.05)
  - method (can be size_cell, insert_buffer…)
  - cell_type (can be combinational, sequential or clock_network)

Write_changes

- After ECO, write_changes command can be applied to PrimeTime to write out eco script for PnR tools

- write_changes \
  - format \
    - ptch (PT script)
    - text (text description of eco)
    - dctcl (design_compiler script)
    - icctcl (icc)
    - icc2tcl (icc2)
    - eco (binary script for PT **read_eco_changes**)
    - aprtcl (3rd party tool)
  - output $filename
  - reset



ECO in PnR tools

In ICC2

```
source $DMSA_SCRIPT
connect_pg_net -automatic
legalize_placement
route_eco -reuse_existing_global_route true -reroute
modified_nets_first_than_others
```

In Innovus

```
source $DMSA_SCRIPT
globalNetConnect VDD –type pgpin –pin {VDD} –inst *
globalNetConnect VDD –type pgpin –pin {VSS} –inst *
refine_placement –preserve_routing true
ecoRoute
```

Enable PT Delay Calculation in ICC2

- In Synopsys PnR tool ICC2, you can enable PrimeTime for delay calculation and timing optimization manually

```
# Enable Synopsys CCS receiver capacitance model during timing
analysis
set_app_options -name time.enable_ccs_rcv_cap -value true

# Enable waveform-based analysis using CCS (requires composite
current source timing and CCS noise data libraries)
set_app_options -name time.delay_calc_waveform_analysis_mode -value
full_design

# Enable combined PrimeTime and waveform-based analysis rules
set_app_options -name time.awp_compatibility_mode -value false
```

Innovus Quantus+Tempus Signoff Flow - 1

- Using Quantus in Innovus:
- setExtractRCMode \

-engine postRoute \

-effortLevel signoff \

-coupled true \

-lefTechFileMap ./SCRIPTS/qrclayermap.ccl

- extractRC
- rcOut -rc_corner $corner -spef $spefFile

Innovus Quantus+Tempus Signoff Flow - 2

- setSignoffOptMode (Enable non-default setting on signoff optimization. Ex: fixXtalk, fixMaxTran…)
- signoffTimeDesign (enable Quantus and Tempus in batch mode and generate ECO db and timing reports)
  - reportOnly
- signoffOptDesign
  - setup
  - hold

- The following example shows signoff STA followed by Hold optimization in PBA Setup and Hold mode:

```
source postRoute.enc
setMultiCpuUsage -localCpu 4 -remoteHost 3 -cpuPerRemoteHost 4
setSignoffOptMode -preStaTcl preStaTcl.tcl
setSignoffOptMode -retime aocv_path_slew_propagation -checkType both
signoffTimeDesign
signoffOptDesign -hold
```

Source: Cadence Innovus 2022.2 UG